PRATEEK KUMAR RAJPUT



PhD Candidate, University of Luxembourg
LLM Agents | Code Security | AI for Software Engineering

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Website GitHub LinkedIn Google Scholar

PhD researcher studying the security and reliability of LLM coding agents. Principal investigator of the FNR-funded TRACER project (€116,000), with industry experience building agent systems and fine-tuning language models. Research connects development traces, program analysis, repair memory, and evaluation beyond functional correctness.

EXPERIENCE

University of Luxembourg, SnT

PhD Researcher, Computer Science

Mar 2024 – Present | Luxembourg

Supervisors:
Prof. Tegawendé F. Bissyandé (Google Scholar)
Prof. Jacques Klein (Google Scholar)

- Develop security review and repair workflows for agent-generated code using development history, functional validation, and persistent repair checks.
- Evaluate code LLMs through structural analysis, runtime profiling, and reproducible experiments across commercial and open-source models.

Contraview

Co-Founder

Aug 2023 – Jan 2026 | Remote, United States

- Built LLM agent systems for biotech investment due diligence, combining retrieval-augmented generation with clinical-trial, disease, and drug information.

Huawei

Research Scientist / Data Scientist

May 2022 – Aug 2023 | Moscow, Russia

- Fine-tuned BART/T5 and built Wikidata subgraph retrieval and answer reranking for complex question answering. Improved MINTAKA Hits@1 by 4–6% over the prior state of the art.

FEATURED PUBLICATION

Securing Agent-Generated Code with Its Own Development History

P. K. Rajput, S. Khan, E.-H. Diallo, Y. Song, J. Klein, T. F. Bissyandé.

Preprint, 2026 | Submitted to ICSE SEIP

TRACER uses agent trajectories for security review, validates repairs, and retains repair memory with executable checks for later changes.

- Symbol-level tracking includes the vulnerable symbol in 94.6% of SusVibes patches, compared with 13.4% for the primary scanner warning.
- In a 10-task case study, memory and boundary checks allow nine follow-up changes to succeed without reintroducing the vulnerability.

RESEARCH FUNDING

TRACER

Principal Investigator

Awarded July 2026

€116,000

FNR TREK/JUMP, Call 2026-1
University of Luxembourg

Trace-based Runtime Analysis for Code Evolution Review.

Official award announcement

EDUCATION

University of Luxembourg
PhD in Computer Science
Mar 2024 – Present
Expected completion: Feb 2028

Skolkovo Institute of Science and Technology
MS in Data Science
2021 – 2023 | Grade: 4.8/5

IIT Kanpur
B.Tech, Mechanical Engineering
2013 – 2017

TECHNICAL SKILLS

Python & Machine Learning

Python, PyTorch, Hugging Face and language-model fine-tuning.

LLM Agents

Tool integration, retrieval-augmented generation, repair memory.

Code Analysis & Evaluation

Static and AST analysis, security validation and LLM evaluation.

PUBLICATIONS & MANUSCRIPTS

Trajectory-Level Security Debt in LLM Coding Agents
P. K. Rajput, A. K. Kabore, Y. Song, M. Tessa, T. Malloy, J. Klein, T. F. Bissyandé.
Accepted, Trustworthy Agentic Systems (TAS), AAAI Fall Symposium, 2026

Correctness isn't Efficiency: Runtime Memory Divergence in LLM-Generated Code
P. K. Rajput, Y. Song, A. A. Bonkoungou, I. E. Olatunji, A. K. Kabore, J. Klein, T. F. Bissyandé.
ICSE, Software Engineering in Practice (SEIP), 2026

Evaluation Drift in LLM Personality Induction: Are We Moving the Goalpost?
P. K. Rajput, Y. Song, I. E. Olatunji, J. Klein, T. F. Bissyandé.
Language Resources and Evaluation Conference (LREC), 2026

Metacognitive Reasoning in Energy Based Models using Instance Based Learning Theory
T. Malloy, **P. K. Rajput**, S. L. Nikiema, C. Gonzalez, T. F. Bissyandé.
Accepted, Symposium on Artificial Metacognition (METACOG-26), AAAI Fall Symposium, 2026

Memorization Diagnostics for Code LLMs Should be Scale-Aware
P. K. Rajput, A. A. Bonkoungou, A. E. Djiré, X. Tang, Y. Song, I. E. Olatunji, E.-H. Diallo, J. Klein, T. F. Bissyandé.
Preprint, 2026 | arXiv:2608.12771

Empirical Study for Structured Output Control in LLMs for Software Engineering
Y. Song, **P. Rajput**, T. Sun, S. Ezzini, T. F. Bissyandé, J. Klein.
Preprint, 2026 | arXiv:2606.09395

Measuring LLM Code Generation Stability via Structural Entropy
Y. Song, T. Sun, X. Tang, **P. K. Rajput**, T. F. Bissyandé, J. Klein.
Automated Software Engineering (ASE), NIER Track, 2025

A Pipeline for Assessing Metacognitive Reasoning in Energy-Based Transformers while Generating Code
T. Malloy, **P. K. Rajput**, I. E. Olatunji, J. Klein, T. F. Bissyandé.
EurIPS Workshop on Metacognition in Generative AI, 2025

Dynamic Stability of LLM-Generated Code
P. Rajput, A. A. Bonkoungou, Y. Song, A. K. Kabore, I. E. Olatunji, J. Klein, T. F. Bissyandé.
Preprint, 2025 | arXiv:2511.07463

Large Language Models Meet Knowledge Graphs to Answer Factoid Questions
M. Salnikov, H. Le, **P. Rajput**, I. Nikishina, P. Braslavski, V. Malykh, A. Panchenko.
Pacific Asia Conference on Language, Information and Computation (PACLIC), 2023

ADDITIONAL TRAINING

Oxford Machine Learning Summer School (OxML 2024) Oxford, UK
Representation Learning & Generative AI, ML for Health & Bio. 44 hours.